

Greatest Common Divisor (GCD) computation

Module I

Algorithm

$$\gcd(a, b) = \begin{cases} a, & \text{if } b = 0 \\ \gcd(b, a \bmod b) & \text{otherwise} \end{cases}$$

Example1: $\gcd(15, 45)$?

- $\gcd(15, 45) = \gcd(45, 15 \bmod 45) = \gcd(45, 15)$
- $\gcd(45, 15) = \gcd(15, 45 \bmod 15) = \gcd(15, 0) = 15$

Example2: $\gcd(120, 295)$?

- $\gcd(295, 120 \bmod 295) = \gcd(295, 120)$
- $\gcd(295, 120) = \gcd(120, 295 \bmod 120) = \gcd(120, 55)$
- $\gcd(120, 55) = \gcd(55, 120 \bmod 55) = \gcd(55, 10)$
- $\gcd(55, 10) = \gcd(10, 55 \bmod 10) = \gcd(10, 5)$
- $\gcd(10, 5) = \gcd(5, 10 \bmod 5) = \gcd(5, 0) = 5$

Algorithm

Example3: gcd (120, 295) ? gcd(n,a), a=295 and n=120.

$\frac{a}{n}, a = nq + r$ from division algorithm, $\frac{295}{120}$.

$$295 = 120 * 2 + 55$$

$$120 = 55 * 2 + 10$$

$$55 = 10 * 5 + 5 \text{ (gcd)}$$

$$10 = 5 * 2 + 0$$

Example4: gcd (1970, 1066) ?